

UNIT-I

SEMICONDUCTOR DIODES AND APPLICATIONS

The p-n junction diode, Equivalent circuit of a diode, Zener diode, Zener diode as a voltage regulator, Rectification - Half wave rectifier, Full wave rectifier, Bridge rectifier, capacitor and choke filter circuit (only qualitative approach), 78XX based Fixed IC voltage regulator.

Formation of PN junction

A PN-junction is formed when an N-type material is fused together with a P-type material creating a semiconductor diode. When the N-type semiconductor and P-type semiconductor materials are first joined together a very large density gradient exists between both sides of the PN junction. The result is that some of the free electrons from the N-side begin to move across this newly formed junction to recombine with holes in the P-side. Similarly holes from the P-side begin to move across the junction to recombine with electrons in the N-side as shown in Fig.1.

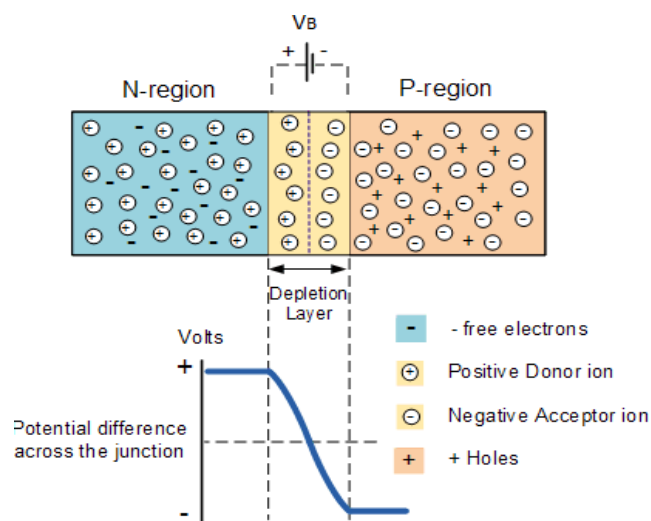


Fig.1 Formation of PN junction

PN Junction Diode Biasing

PN Junction Diode can be biased by an external voltage to either block or allow the flow of current through it.

Forward Bias

When positive terminal is connected to the P-type material and negative to the N-type material of the diode as shown in Fig.2, it is known as forward bias.

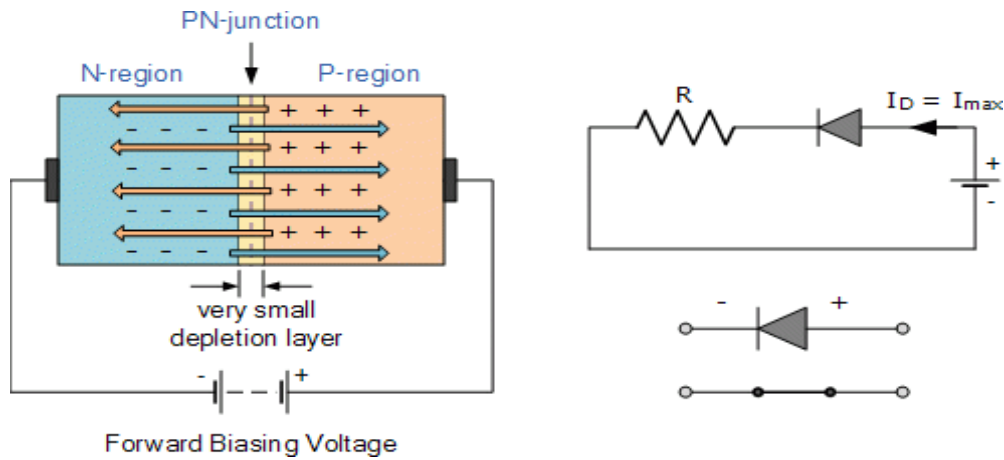


Fig.2 : Forward biased pn-junction diode

The effect of adding this additional energy source results in the electrons and holes being able to cross the depletion region from one side to the other. If this external voltage becomes greater than the value of the potential barrier, approx. 0.7 V for silicon and 0.3 V for germanium diodes, the holes from p-side and electrons from n-side get pushed towards the junction and current starts flowing. This voltage is called the “knee voltage”. After this, current increases with little increase in the external voltage as shown in the static V-I characteristics.

Reverse Bias

When negative terminal is connected to the P-type material and positive to the N-type material of the diode as shown in Fig.3, it is known as reverse bias.

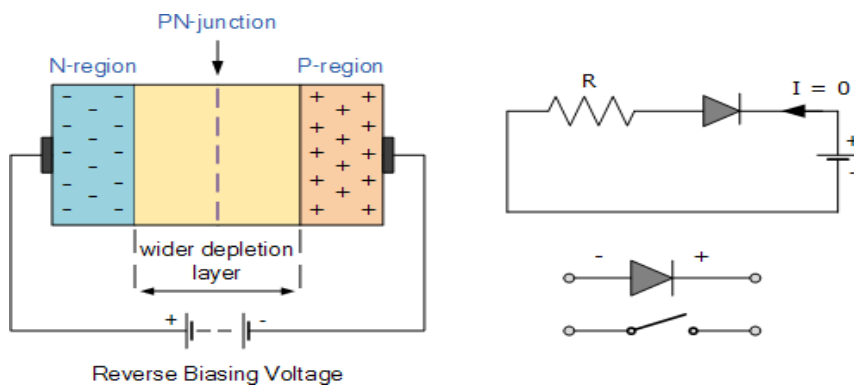


Fig.3: Reverse biased pn-junction diode

The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the P-type end are also attracted away from the junction towards the negative electrode. The net result is that the depletion layer grows wider due to a lack of electrons and holes and presents a high impedance path, almost an insulator. The result is that a high potential barrier is created thus preventing current from flowing through the semiconductor material. This condition represents a high resistance value to the PN junction and practically zero current flows through the junction diode with an increase in bias voltage. However, a very small leakage current does flow through the junction due to the minority electrons in P-side and electrons in N-side, which is measured in μA for germanium and nA for silicon.

If the reverse bias voltage V_R applied to the diode is increased to a sufficiently high value, PN junction will overheat and fail due to the *avalanche effect* around the junction. This may cause the diode to become shorted and will result in the flow of maximum circuit current which is shown as a step downward slope in the static V-I characteristics.

Diode Symbol and Static V-I Characteristics

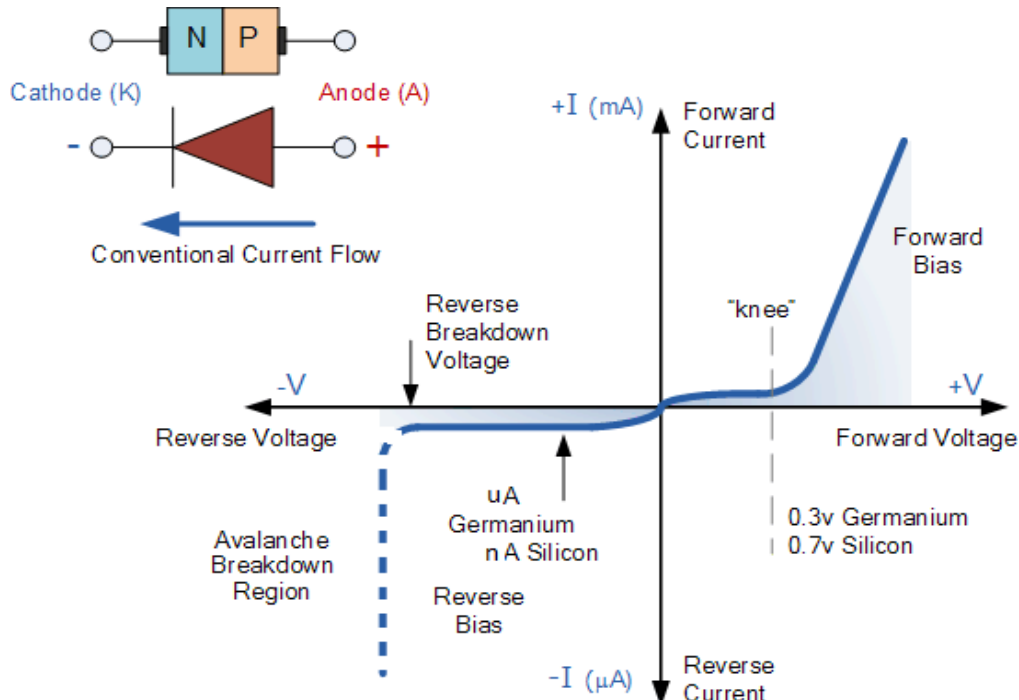


Fig.4: Diode static V-I characteristics

There are two operating regions and three possible “biasing” conditions for the standard **Junction Diode** and these are: Forward Bias and Reverse Bias.

When the diode is forward biased “Forward Current” I_F flows. Before the knee voltage the forward current is due to the diffusion of holes and electrons across the junction. After the knee voltage the current is mainly due to the drift of charge carriers. The potential difference or voltage across the diode is constant when it conducts forward current I_F and is approximately 0.3 V for germanium and approximately 0.7 V for silicon diode. The (slope)⁻¹ of the forward characteristics after the knee of the characteristics is the forward resistance of the diode when it is conducting as shown in Fig.4.

“Reverse Current” which is referred as I_R is the reverse saturation current which flows due to the minority carriers. The minority carriers are constantly generated due to thermal energy raising the temperature of the PN junction which in turn causing an increase in the generation of minority carriers, thereby resulting in an increase in leakage current. When the reverse voltage applied equals the reverse breakdown voltage V_{BR} a large reverse current flows destroying the PN junction. The resistors are used in series with the diode to limit its current flow. Exceeding its maximum forward current specification causes the device to dissipate more power in the form of heat.

Diodes are manufactured in a range of voltage and current ratings and care must be taken when choosing a diode for a certain application.

Some of the parameters of pn-diode are explained as follows

1. Maximum Forward Current (I_{Fmax})

The Maximum Forward Current I_{Fmax} is as its name implies the maximum forward current allowed to flow through the device. When the diode is conducting in the forward bias condition, it has a very small “ON” resistance across the PN junction and therefore, power is dissipated across this junction in the form of heat.

2. Forward Voltage drop/ Knee Voltage (V_F)

It is a forward voltage drop across a diode when it conducts. The forward bias voltage must exceed this voltage for the diode to conduct. Forward voltage drop for silicon diodes is 0.7V and for germanium diodes is 0.3 V.

3. Reverse saturation current (I_R)

It is the nominal current which flows through the diode when it is reverse biased. It is in the order of nA for Si diodes and μA for Ge diodes.

4. Reverse breakdown voltage (V_{BR})

It is the reverse bias voltage at which the pn-junction breaks down and permanently damages the diode. V_{BR} is around 50V for Si diode and 65 V for Ge.

5. Dynamic resistance (r_d)

The dynamic resistance or ac resistance of a diode is the reciprocal of the slope of the forward characteristics beyond knee.

$$r_d = \frac{\Delta V_F}{\Delta I_F}$$

Diode Models / Diode Equivalent Circuits

The three models of diode are as shown in Fig.5. The diode is two terminal non-linear device whose V-I characteristics are dependent on the polarity of the applied voltage. The diode is *Forward Biased* for $V_D > 0$, *Reverse Biased* for $V_D < 0$.

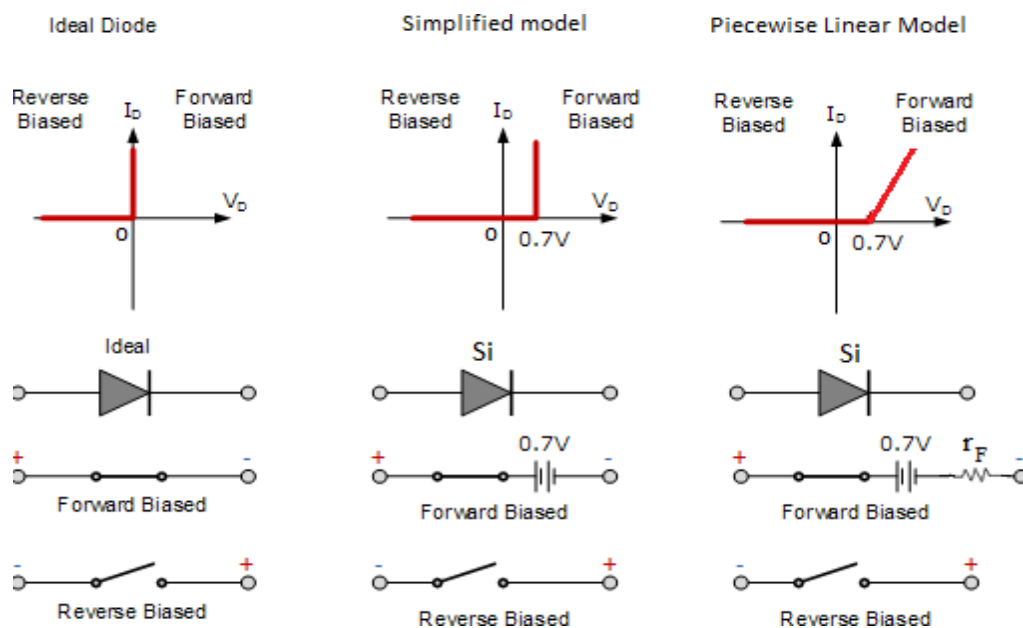


Fig.5 : Diode models

DC load line analysis

A DC load line is a straight line drawn on the diode forward characteristics that describes all the dc conditions that exist in the operation of the circuit. Consider a diode series circuit as shown in Fig.6.

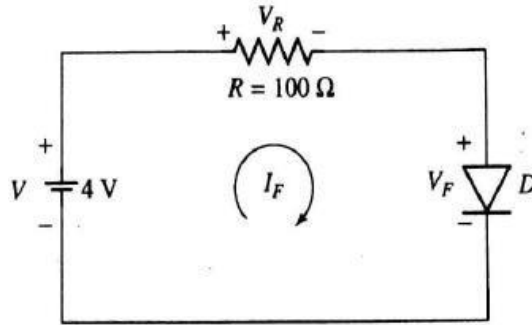


Fig.6: Diode series circuit

Here I_F is the current through the diode and V_F is the voltage across the diode. Applying KVL to the circuit we get,

$$\begin{aligned} -V + V_R + V_F &= 0 \\ \text{or } V &= I_F R + V_F \end{aligned} \quad (1)$$

Eq.1 is the equation of load line.

For $I_F = 0$, $V_F = V$. Hence when $I_F = 0$, $V_F = V$ is one of the points on load line at point A.

$$V_F = 4V, I_F = 0 \text{ mA}$$

From eq. 1, when $V_F = 0$, $I_F = \frac{V}{R}$.

$V_F = 0, I_F = \frac{V}{R}$ is the second point on the DC load line at point B.

$V_F = 0, I_F = \frac{4}{100} = 40 \text{ mA}$. Join the points A and B to get the DC load line. We can see that the DC load line intersects the diode characteristics at a point Q which corresponds to

$I_F = 33 \text{ mA}$ and $V_F = 0.7V$. This can be shown by applying KVL to the circuit as

$$V - V_F - I_F R = 0 \quad \text{or}$$

$$I_F = \frac{V - V_F}{R}$$

$V_F = 0.7 \text{ V}$ for Silicon diode

$$\therefore I_F = \frac{4 - 0.7}{100} = 33 \text{ mA}$$

The DC load line is drawn on the diode forward characteristics is as shown in Fig.7.

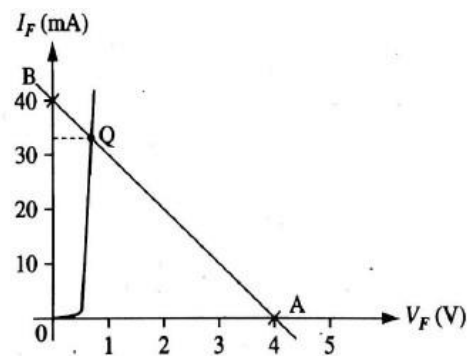


Fig.7: Plot of DC load line.

Rectifiers using Diodes

Half wave rectifier, Full wave rectifier, Bridge rectifier, capacitor and choke filter circuit (only qualitative approach)

A rectifier is a circuit which converts the *Alternating Current* (AC) input power into a *Direct Current* (DC) output power. The input power supply may be either a single-phase or a multi-phase supply.

Half Wave Rectifier.

The simplest of all the rectifier circuits is Half Wave Rectifier. In a half wave rectifier circuit one half of each complete sine wave of the AC supply is converted to DC. That's why this type of circuit is called a "half-wave" rectifier because it passes only half of the incoming AC power supply as shown in Fig.8.

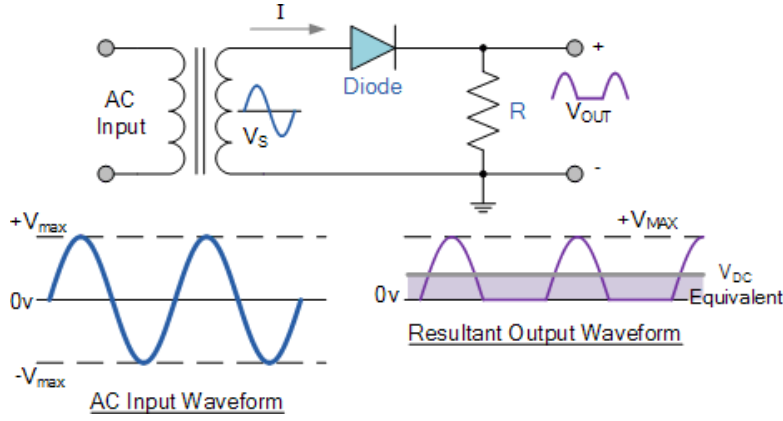


Fig.8: Half Wave Rectifier

During each “positive” half cycle of the AC sine wave, the diode is *forward biased* as the anode is positive with respect to the cathode resulting in current flowing through the diode. The voltage across the load resistor will therefore be the same as the supply voltage, V_s ($V_s - V_F$ for Si or Ge diode), that is the “DC” voltage across the load is sinusoidal for the first half cycle only so $V_{out} = V_s$.

During each “negative” half cycle of the AC sinusoidal input waveform, the diode is *reverse biased* as the anode is negative with respect to the cathode. Therefore, NO current flows through the diode or circuit. Then in the negative half cycle of the supply, no current flows in the load resistor as no voltage appears across it so therefore, $V_{out} = 0$.

The current on the DC side of the circuit flows in one direction only making the circuit **Unidirectional**. As the load resistor receives from the diode a positive half of the waveform, zero volts, a positive half of the waveform, zero volts, etc.

Expressions for V_{DC} , V_{rms} and Rectification Efficiency, Ripple factor for Half Wave Rectifier

Consider output voltage,

$$V_o = V_m \sin \omega t \text{ for } 0 \leq \omega t \leq \pi$$

$$V_o = 0 \text{ for } \pi \leq \omega t \leq 2\pi$$

V_i is a sinusoidal waveform it can be represented as,

$$V_i = V_m \sin \omega t = V_m \sin \omega t$$

Load current,

$$I_L = \begin{cases} I_m \sin \omega t & \text{for } 0 \leq \omega t \leq \pi \\ 0 & \pi \leq \omega t \leq 2\pi \end{cases}$$

Where Max. load current, $I_m = \frac{V_m}{R_f + R_s + R_L}$

where R_f is the dynamic resistance of the diode

R_L is the load resistance & R_s transformer secondary winding resistance.

Derivation of V_{DC}

DC Average current I_{DC}

$$I_{DC} = \frac{\text{Area under curve over full cycle}}{\text{Time period}} = \frac{1}{2\pi} \int_0^{2\pi} I_m \sin \omega t d\omega t$$

$$I_{DC} = \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t + \int_{\pi}^{2\pi} I_m \sin \omega t d\omega t \right]$$

$$I_{DC} = \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t + \int_{\pi}^{2\pi} 0 d\omega t \right] \text{ since } I_L = 0, \pi \leq \omega t \leq 2\pi$$

$$I_{DC} = \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin \omega t d\omega t \right]$$

$$I_{DC} = \frac{I_m}{2\pi} \left[-\cos \omega t \right]_0^{\pi} = \frac{-I_m}{2\pi} [-1 - 1] = \frac{I_m}{\pi}$$

DC Average Voltage V_{DC}

$$V_{DC} = I_{DC} R_L = \frac{I_m}{\pi} R_L$$

$$\text{Where } I_m = \frac{V_m}{R_f + R_s + R_L}$$

$$\square V_{DC} = \frac{V_m}{R_f + R_s + R_L} \frac{R_L}{\pi} = \frac{V_m}{\pi} \frac{R_L}{R_f + R_s + R_L}$$

When the load resistor $R_L \gg R_f$ and R_s ,

The dc value of the output voltage is $V_{dc} = \frac{V_m}{\pi}$

Derivation of V_{rms}

Root mean square load current, I_{rms}

$$\begin{aligned}
 I_{rms} &= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_L^2 d\omega t} = \sqrt{\frac{1}{2\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t} \\
 I_{rms} &= \sqrt{\frac{1}{2\pi} \left[\int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t + \int_{\pi}^{2\pi} I_m^2 \sin^2 \omega t d\omega t \right]} \\
 &= \sqrt{\frac{1}{2\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t} \quad \text{since } I_L = 0, \pi \leq \omega t \leq 2\pi \\
 &= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} \frac{1 - \cos 2\omega t}{2} d\omega t} = \sqrt{\frac{I_m^2}{2\pi} \left[\frac{1}{2} \int_0^{\pi} d\omega t - \frac{1}{2} \int_0^{\pi} \cos 2\omega t d\omega t \right]} \\
 &= \sqrt{\frac{I_m^2}{2\pi} \left[\frac{1}{2} \left[\omega t \right]_0^{\pi} - \frac{1}{2} \left[\frac{\sin 2\omega t}{2} \right]_0^{\pi} \right]} \\
 &= \sqrt{\frac{I_m^2}{2\pi} \left[\frac{\pi}{2} - 0 \right]} = \frac{I_m}{2}
 \end{aligned}$$

Root mean square load voltage, V_{rms}

$$V_{rms} = I_{rms} R_L = \frac{I_m}{2} R_L$$

$$\text{Where } I_m = \frac{V_m}{R_f + R_s + R_L}$$

$$\square V_{rms} = \frac{V_m R_L}{R_f + R_s + R_L} \cdot \frac{1}{2} = \frac{V_m R_L}{2(R_f + R_s + R_L)}$$

When the load resistor $R_L \gg R_f$ and R_s ,

The rms value of the output voltage is $V_{rms} = \frac{V_m}{2}$

Derivation of Ripple factor

The ripple factor is the measure of conversion from AC to DC of the half wave rectifier is

$$r = \frac{\text{rms value of ac component of the output}}{\text{dc value of the output}} = \frac{I_{ac}}{I_{dc}} \quad \text{---(1)}$$

$$\text{Where } I_{ac}^2 = I_{rms}^2 - I_{dc}^2$$

Substituting in (1),

$$r = \frac{I_{ac}}{I_{dc}} = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}} = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$\text{Substitute, } I_{rms} = \frac{I_m}{2} \text{ \& } I_{dc} = \frac{I_m}{\pi}$$

$$\text{We get, } r = \sqrt{\left(\frac{\frac{I_m}{2}}{\frac{I_m}{\pi}}\right)^2 - 1} = \mathbf{1.21}$$

Thus, the ripple factor of the half wave rectifier is 1.21 or the amount of ac present in the output is 121 % of the dc voltage.

Derivation of Rectification Efficiency

The efficiency of the half wave rectifier is

$$\eta = \frac{\text{dc power delivered to the load}}{\text{ac input power from the transformer secondary}} = \frac{P_{dc}}{P_{ac}}$$

$$\text{DC output power } P_{dc} = I_{dc}^2 R_L$$

$$\text{AC input power } P_{ac} = I_{rms}^2 (R_f + R_s + R_L)$$

$$\eta = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_f + R_s + R_L)}$$

$$\text{Substitute, } I_{rms} = \frac{I_m}{2} \text{ \& } I_{dc} = \frac{I_m}{\pi}$$

$$\square \eta = \frac{\left(\frac{I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{2}\right)^2 (R_f + R_s + R_L)} = \mathbf{0.406 = 40.6\%}$$

If $R_L \gg R_f$, the efficiency $\eta = \mathbf{40.6\%}$ or with no diode loss only 40.6% of the ac input power is converted into dc power in the load.

During the rectification process, the resultant output DC voltage and current are therefore both “ON” and “OFF” during every cycle. As the voltage across the load resistor is only present during the positive half of the cycle (50% of the input waveform), this results in a low average DC value being supplied to the load.

The variation of the rectified output waveform between this “ON” and “OFF” condition produces a waveform which has large amounts of “ripple” which is an undesirable feature. The resultant DC ripple has a frequency that is equal to that of the AC supply frequency.

Full Wave Rectifier.

Like the half wave circuit, a full wave rectifier circuit produces an output voltage or current which is purely DC or has some specified DC component. Full wave rectifiers have some fundamental advantages over the half wave rectifier counterparts. The average (DC) output voltage is higher than for half wave, the output of the full wave rectifier has much less ripple than that of the half wave rectifier producing a smoother output waveform.

Full Wave Bridge Rectifier

Another type of circuit that produces the same output waveform as the center tapped transformer rectifier is the **Full Wave Bridge Rectifier**. This type of single phase rectifier uses four individual rectifying diodes connected in a closed loop “bridge” configuration to produce the desired output.

The main advantage of this bridge circuit is that it does not require a special centre tapped transformer, thereby reducing its size and cost. The single secondary winding is connected to one side of the diode bridge network and the load to the other side as shown in Fig.10.

The four diodes labelled D_1 to D_4 are arranged in “series pairs” with only two diodes conducting current during each half cycle.

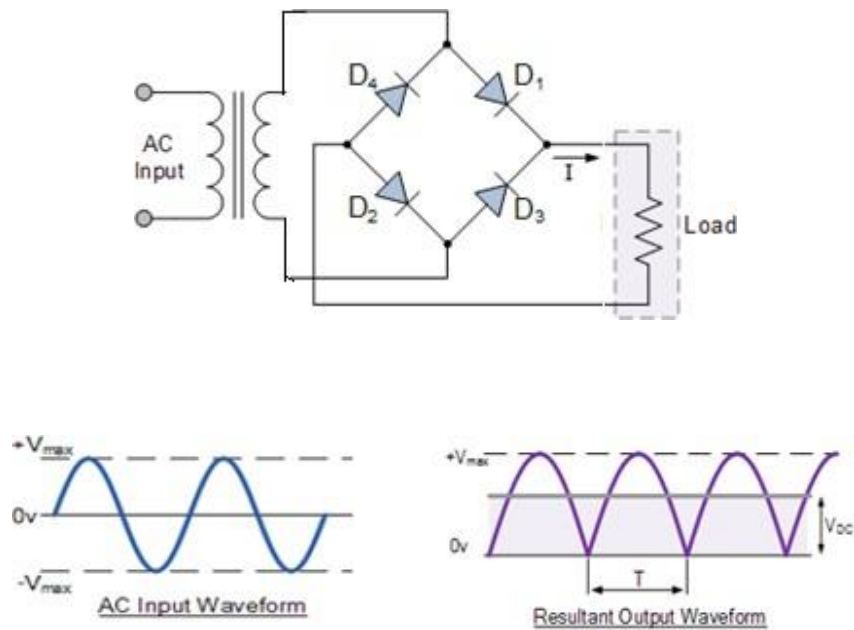


Fig.10: Bridge Rectifier

During the positive half cycle of the supply, diodes D_1 and D_2 conduct in series while diodes D_3 and D_4 are reverse biased and the current flows through the load as shown in the Fig.11.

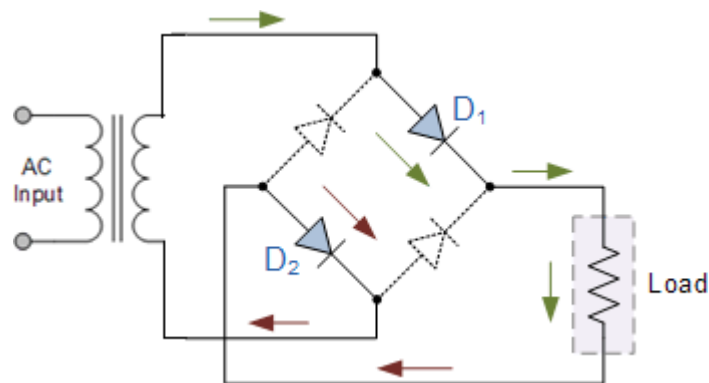


Fig.11: Working of Bridge Rectifier for Input Positive halfcycle

During the negative half cycle of the supply, diodes D_3 and D_4 conduct in series, but diodes D_1 and D_2 switch “OFF” as they are now reverse biased. The current flowing through the load is the same direction as before as shown in the Fig.12.

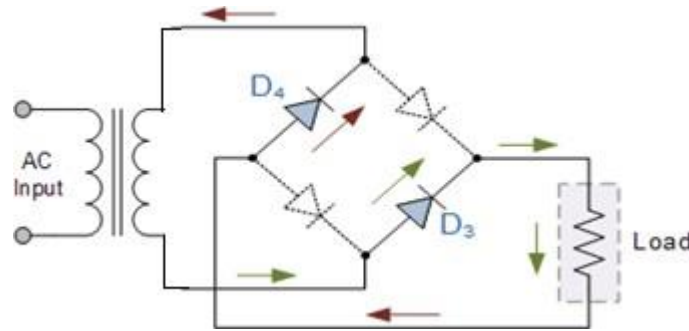


Fig.12: Working of Bridge Rectifier for Input Negative halfcycle

As the current flowing through the load is unidirectional, so the voltage developed across the load is also unidirectional and same as the previous two diode full-wave rectifier.

However in reality, during each half cycle the current flows through two diodes instead of just one so the amplitude of the output voltage is two voltage drops ($2 \times 0.7 = 1.4\text{V}$) less than the input V_{MAX} amplitude. The ripple frequency is now twice the supply frequency (e.g. 100Hz for a 50Hz supply or 120Hz for a 60Hz supply.)

Although we can use four individual power diodes to make a full wave bridge rectifier, pre-made bridge rectifier components are available “off-the-shelf” in a range of different voltage and current sizes that can be soldered directly into a PCB circuit board or be connected by spade connectors.

Expressions for V_{dc} , V_{rms} , Ripple factor and Rectification efficiency for Full Wave Rectifier

Derivation of V_{DC}

DC Average current I_{DC}

$$I_{\text{DC}} = \frac{\text{Area under curve over full cycle}}{\text{Time period}} = \frac{1}{2\pi} \int_0^{2\pi} I_m \sin\omega t d\omega t$$

$$I_{\text{DC}} = \frac{1}{2\pi} \left[\int_0^{\pi} I_m \sin\omega t d\omega t + \int_{\pi}^{2\pi} I_m \sin\omega t d\omega t \right]$$

$$I_{\text{DC}} = \frac{1}{2\pi} \left[2 \int_0^{\pi} I_m \sin\omega t d\omega t \right] = \frac{1}{\pi} \left[\int_0^{\pi} I_m \sin\omega t d\omega t \right]$$

$$I_{\text{DC}} = \frac{I_m}{\pi} \left[-\cos\omega t \right]_0^{\pi} = \frac{-I_m}{2\pi} [-1-1] = \frac{2I_m}{\pi}$$

DC Average Voltage V_{DC}

$$V_{DC} = I_{DC} R_L = \frac{2I_m}{\pi} R_L$$

$$\text{Where } I_m = \frac{V_m}{R_f + R_s + R_L}$$

$$\square V_{DC} = \frac{2V_m}{\pi} \frac{R_L}{R_f + R_s + R_L} = \frac{2V_m}{\pi} \frac{R_L}{R_f + R_s + R_L}$$

When the load resistor $R_L \gg R_f$ and R_s ,

The dc value of the output voltage is $V_{dc} = \frac{2V_m}{\pi}$

where R_s is the series resistance of the source V_s

R_f is the dynamic resistance of the diode

R_L is the load resistance.

Derivation of V_{rms}

Root mean square load current, I_{rms}

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_L^2 d\omega t} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_m^2 \sin^2 \omega t d\omega t} \\ I_{rms} &= \sqrt{\frac{1}{2\pi} \left[\int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t + \int_{\pi}^{2\pi} I_m^2 \sin^2 \omega t d\omega t \right]} \\ &= \sqrt{\frac{1}{2\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t d\omega t} \\ &= \sqrt{\frac{I_m^2}{\pi} \int_0^{\pi} \frac{1 - \cos 2\omega t}{2} d\omega t} = \sqrt{\frac{I_m^2}{\pi} \left[\frac{1}{2} \int_0^{\pi} d\omega t - \frac{1}{2} \int_0^{\pi} \cos 2\omega t d\omega t \right]} \\ &= \sqrt{\frac{I_m^2}{\pi} \left[\frac{1}{2} \left[\omega t \right]_0^{\pi} - \frac{1}{2} \left[\frac{\sin 2\omega t}{2} \right]_0^{\pi} \right]} = \sqrt{\frac{I_m^2}{\pi} \left[\frac{\pi}{2} - 0 \right]} = \frac{I_m}{\sqrt{2}} \end{aligned}$$

Root mean square load voltage, V_{rms}

$$V_{rms} = I_{rms} R_L = \frac{I_m}{\sqrt{2}} R_L$$

Where , $I_m = \frac{V_m}{R_f + R_S + R_L}$

$$\square V_{rms} = \frac{V_m}{R_f + R_S + R_L} \frac{R_L}{\sqrt{2}} = \frac{V_m}{\sqrt{2}} \frac{R_L}{R_f + R_S + R_L}$$

When the load resistor $R_L \gg R_f$ and R_S ,

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

Derivation of Ripple factor

The ripple factor, the measure of conversion from AC to DC of the full wave rectifier is

$$r = \frac{\text{rms value of ac component of the output}}{\text{dc value of the output}} = \frac{I_{ac}}{I_{dc}}$$

$$\text{where } I_{ac}^2 = I_{rms}^2 - I_{dc}^2$$

$$r = \frac{I_{ac}}{I_{dc}} = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}} = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$\text{Substitute, } I_{rms} = \frac{I_m}{\sqrt{2}} \text{ \& } I_{dc} = \frac{2I_m}{\pi}$$

$$\text{We get, } r = \sqrt{\left(\frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}}\right)^2 - 1} = 0.482$$

Thus, the ripple factor of the full wave rectifier is 0.482 or the amount of ac present in the output is 48.2 % of the dc voltage, which is a drastic improvement from 121 % of half wave rectifier.

Derivation of Rectification Efficiency

The efficiency of the full wave rectifier is

$$\% \eta = \frac{\text{dc power delivered to the load}}{\text{ac input power from the transformer secondary}} \times 100 = \frac{P_{dc}}{P_{ac}} \times 100$$

$$\text{DC output power } P_{dc} = I_{dc}^2 R_L$$

AC input power $P_{ac} = I_{rms}^2(R_f + R_s + R_L)$

$$\eta = \frac{I_{dc}^2 R_L}{I_{rms}^2(R_f + R_s + R_L)}$$

$$\text{Substitute, } I_{rms} = \frac{I_m}{\sqrt{2}} \text{ \& } I_{dc} = \frac{2I_m}{\pi}$$

$$\text{we get } \eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (R_f + R_s + R_L)} = \mathbf{0.812 = 81.2 \%}$$

If $R_L \gg R_s + R_f$, the efficiency $\gamma = \mathbf{81.2 \%}$ or with no diode loss 81.2 % of the ac input power is converted into dc power in the load.

The main disadvantage of this type of full wave rectifier circuit is that a larger transformer for a given power output is required with two separate but identical secondary windings making this type of full wave rectifying circuit costly compared to the “Full Wave Bridge Rectifier” circuit equivalent.

Full-wave Rectifier with Capacitor filter

We saw in the previous section that the single phase half-wave rectifier produces an output wave every half cycle and that it was not practical to use this type of circuit to produce a steady DC supply. The full-wave bridge rectifier however, gives us a greater mean DC value (0.637 Vmax) with less superimposed ripple while the output waveform is twice that of the frequency of the input supply frequency.

The average DC output of the rectifier can be improved while at the same time reducing the AC variation of the rectified output by using smoothing capacitors to filter the output waveform. Smoothing or reservoir capacitors connected in parallel with the load across the output of the full wave bridge rectifier circuit increases the average DC output level even higher as the capacitor acts like a storage device as shown in Fig.13.

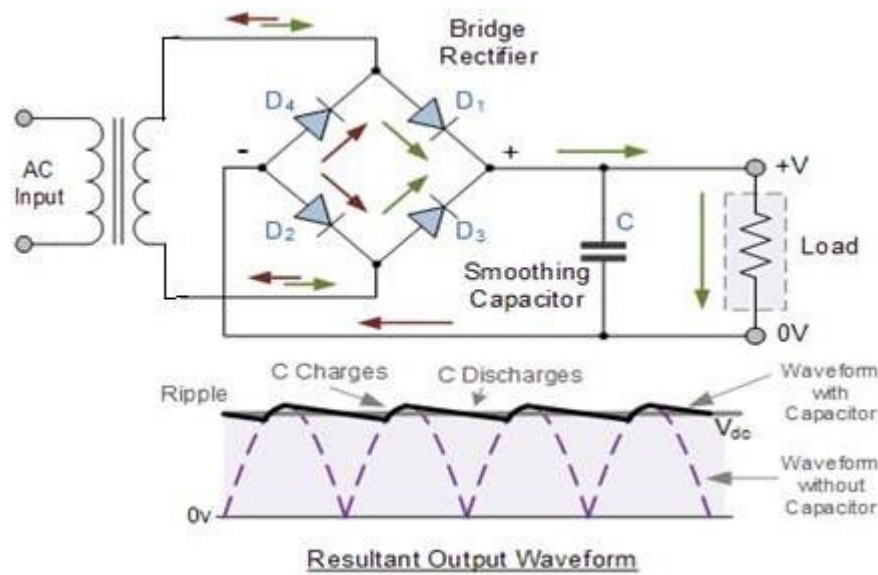


Fig13: Bridge rectifier with capacitor filter

The smoothing capacitor converts the full-wave rippled output of the rectifier into a more smooth DC output voltage. Here the capacitor is charged to the peak voltage of the output DC output, but when it drops from its peak voltage back down to zero volts, the capacitor cannot discharge as quickly due to the RC time constant of the circuit, maintaining the voltage across the load resistor until the capacitor re-charges once again on the next positive slope of the DC output. In other words, the capacitor only has time to discharge briefly before the next DC pulse recharges it back up to the peak value. Thus, the DC voltage applied to the load resistor drops only by a small amount. But we can improve this still by increasing the value of the smoothing capacitor.

Choke capacitor filter

It consists of an inductor connected in series with rectifier output circuit and a capacitor connected in parallel with the load resistor R_L as in Fig14.

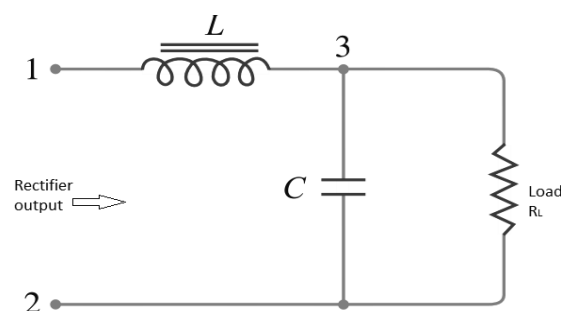


Fig.14: Choke capacitor filter

When the pulsating DC signal from the output of the rectifier circuit is fed into choke filter, the AC ripples present in the output DC voltage gets filtered by choke coil. The inductor has the property to block AC and pass DC. This is because DC resistance of an inductor is low and AC impedance of inductor coil is high. Thus, the AC ripples get blocked by inductor coil.

Although the inductor efficiently removes AC ripples, a small percentage of AC ripples is still present in the filtered signal. These ripples are then removed by the capacitor connected in parallel to the load resistor. Now, the DC output signal is free from AC components, and this regulated DC can be used in any application.

Zener Diode

The **Zener Diode** or “Breakdown Diode”, as they are sometimes referred too, are basically the same as the standard PN junction diode but they are specially designed to have a low and specified **Reverse Breakdown Voltage** which takes advantage of any reverse voltage applied to it.

The **Zener diode** behaves just like a normal general-purpose diode consisting of a silicon PN junction and when biased in the forward direction, that is Anode positive with respect to its Cathode, it behaves just like a normal signal diode passing the rated current.

However, unlike a conventional diode that blocks any flow of current through itself when reverse biased, that is the Cathode becomes more positive than the Anode, as soon as the reverse voltage reaches a pre-determined value, the zener diode begins to conduct in the reverse direction.

This is because when the reverse voltage applied across the zener diode exceeds the rated voltage of the device a process called *Avalanche Breakdown* occurs in the semiconductor depletion layer and a current starts to flow through the diode to limit this increase in voltage.

The current now flowing through the zener diode increases dramatically to the maximum circuit value (which is usually limited by a series resistor) and once achieved, this reverse saturation current remains fairly constant over a wide range of reverse voltages. The voltage point at which the voltage across the zener diode becomes stable is called the “zener voltage”, (V_z) and for zener diodes this voltage can range from less than one volt to a few hundred volts.

The point at which the zener voltage triggers the current to flow through the diode can be very accurately controlled (to less than 1% tolerance) in the doping stage of the diodes semiconductor construction giving the diode a specific *zener breakdown voltage*, (V_Z) for example, 4.3V or 7.5V. This zener breakdown voltage on the I-V curve is almost a vertical straight line.

Zener Diode V-I Characteristics

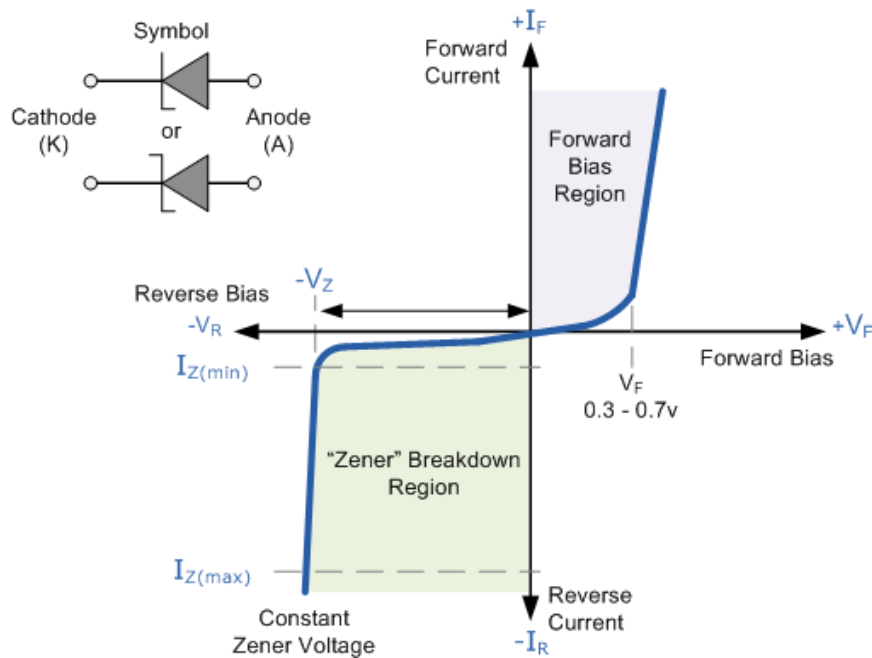


Fig.15: Zener Diode V-I Characteristics

The **Zener Diode** is used in its “reverse bias” or reverse breakdown mode, i.e. the diodes anode connects to the negative supply. From the I-V characteristics curve above, as seen in Fig.15, the zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode and remains nearly constant even with large changes in current as long as the zener diodes current remains between the breakdown current $I_{Z(min)}$ and the maximum current rating $I_{Z(max)}$.

This ability to control itself can be used to great effect to regulate or stabilise a voltage source against supply or load variations. The fact that the voltage across the diode in the breakdown region is almost constant turns out to be an important characteristic of the zener diode as it can be used in the simplest types of voltage regulator applications.

The function of a regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or the variation in the load current and the zener diode will continue to regulate the voltage until the diodes current falls below the minimum $I_{Z(\min)}$ value in the reverse breakdown region.

The Zener Diode Regulator

Zener Diodes can be used to produce a stabilised voltage output with low ripple under varying load current conditions. By passing a small current through the diode from a voltage source, via a suitable current limiting resistor (R_S), the zener diode will conduct sufficient current to maintain a voltage drop of V_{out} .

We remember from the previous tutorials that the DC output voltage from the half or full-wave rectifiers contains ripple superimposed onto the DC voltage and that as the load value changes so to does the average output voltage. By connecting a simple zener stabiliser circuit as shown below across the output of the rectifier, a more stable output voltage can be produced.

Zener Diode Voltage Regulator

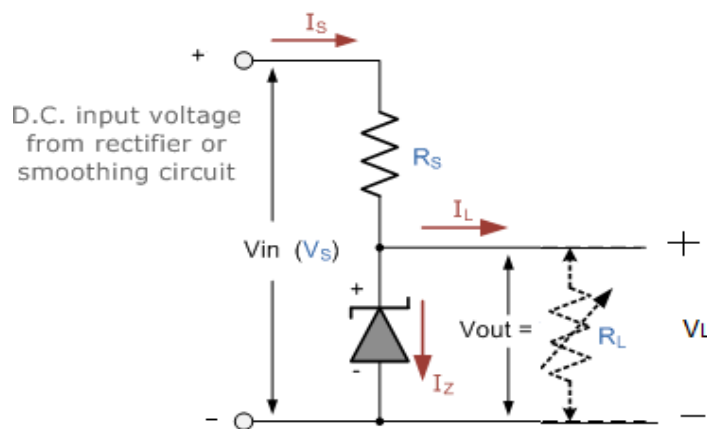


Fig.16: Zener Diode Voltage Regulator

The resistor, R_S is connected in series with the zener diode to limit the current flow through the diode with the voltage source, V_S being connected across the combination. The stabilised output voltage V_{out} is taken from across the zener diode. The zener diode is connected with its cathode terminal connected to the positive side of the DC supply so it is reverse biased and

will be operating in its breakdown condition. Resistor R_S is selected so to limit the maximum current flowing in the circuit.

When no load is connected ($I_L = 0$): With no load connected to the circuit, the load current will be zero, ($I_L = 0$), and all the circuit current passes through the zener diode which in turn dissipates its maximum power. Also a small value of the series resistor R_S will result in a greater diode current when the load resistance R_L is connected and large as this will increase the power dissipation requirement of the diode so care must be taken when selecting the appropriate value of series resistance so that the zener's maximum power rating is not exceeded under this no-load or high-impedance condition.

When the load R_L is connected across the diode: The load is connected in parallel with the zener diode, so the voltage across R_L is always the same as the zener voltage, ($V_R = V_Z$). There is a minimum zener current for which the stabilization of the voltage is effective and the zener current must stay above this value operating under load within its breakdown region at all times. The upper limit of current is of course dependant upon the power rating of the device. The supply voltage V_S must be greater than V_Z .

Voltage regulation can be done through two techniques:

1. Line regulation: In this case, series resistor R_S and load resistor R_L are kept constant. It is assumed that all the variations in voltage arise due to fluctuations in input power supply. The regulated output voltage is achieved for input voltage above certain minimum level.

% regulation is given by

$$\frac{\Delta V_o}{\Delta V_{in}} \times 100$$

2. Load regulation: In this case, the input voltage is fixed, while the load resistance is varied. The constant output voltage is obtained as long as the load resistance R_L is maintained above a minimum value.

% regulation is given by $\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$

A zener diode is always operated in its reverse biased condition. A voltage regulator circuit can be designed using a zener diode to maintain a constant DC output voltage across the load in spite of variations in the input voltage or changes in the load current. The zener voltage

regulator consists of a current limiting resistor R_S connected in series with the input voltage V_S with the zener diode connected in parallel with the load R_L in this reverse biased condition. The stabilized output voltage is always selected to be the same as the breakdown voltage V_Z of the diode.

Zener Diode Example 1:

A 5.0V stabilised power supply is required to be produced from a 12V DC power supply input source. The maximum power rating P_Z of the zener diode is 2W. Using the zener regulator circuit in Fig.16, calculate:

a). The maximum current flowing through the zener diode.

$$\text{Maximum Current} = \frac{\text{Watts}}{\text{Voltage}} = \frac{2\text{w}}{5\text{v}} = 400\text{mA}$$

b). The minimum value of the series resistor, R_S

$$R_S = \frac{V_S - V_Z}{I_Z} = \frac{12 - 5}{400\text{mA}} = 17.5\Omega$$

c). The load current I_L if a load resistor of $1\text{k}\Omega$ is connected across the zener diode.

$$I_L = \frac{V_Z}{R_L} = \frac{5\text{v}}{1000\Omega} = 5\text{mA}$$

d). The zener current I_Z at full load.

$$I_Z = I_S - I_L = 400\text{mA} - 5\text{mA} = 395\text{mA}$$

A typical **zener diode** for general electronic circuits is the 500mW, *BZX55* series or the larger 1.3W, *BZX85* series where the zener voltage is given as, for example, *C7V5* for a 7.5V diode giving a diode reference number of *BZX55C7V5*.

BJT and Applications

The most popular semiconductor device – BJT (Bipolar Junction Transistor). The name bipolar is used because both types of carriers namely holes and electrons are used in the transistor, as opposed to field effect transistor, which is considered a unipolar device.

Transistor was invented by J. Bardeen (1908 - 1987), W. Shockley (1910-1989), and W. Brattain (1902 - 1987) in 1948. In 1950 the junction transistor was made using molten germanium. The bipolar transistors produced in 1950s were typically made with alloyed junction.

Most important applications of BJT are – as an amplifier, oscillator and as a switch.

Construction:

BJT is three terminal device – Base, Emitter and Collector

The arrow in the symbol of transistor shows direction of flow of current in an emitter. Bipolar junction transistor, BJT can be viewed as two *pn* junctions connected back to back to form *np-pn* or *pn-np* structures namely as *nnp* or *pnp* transistors.

Fig.1 illustrates the structure showing two *pn* junctions connected back to back.

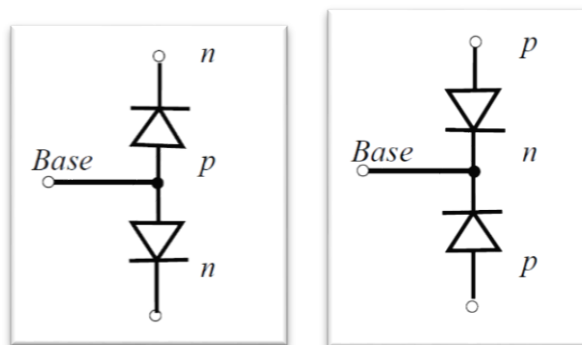


Fig.1: Types of transistors

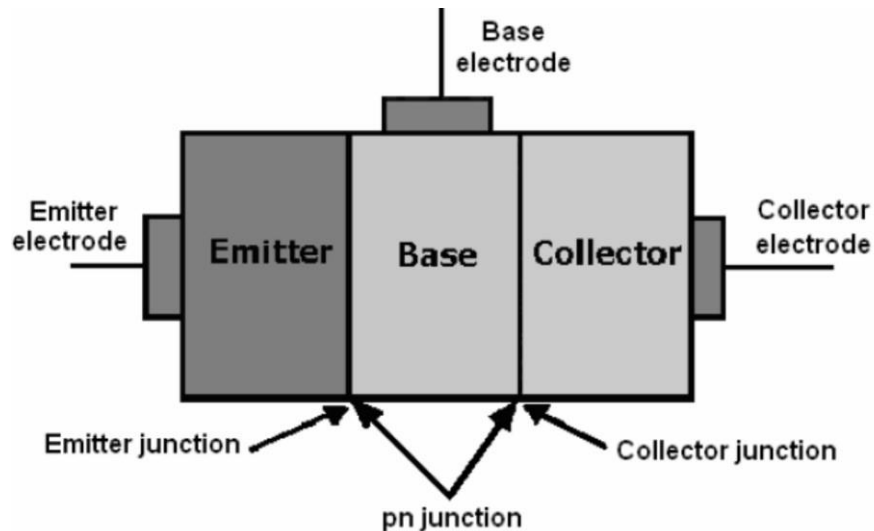


Fig.2: Structure of transistor showing two pn junctions

The transistor consists of two pn junctions, each with its own depletion region as shown in Fig.2 which are explained below

1. Emitter-base junction: Since the emitter is usually heavily doped, the depletion region lies almost entirely in the base.
2. Base-collector junction: The depletion region at this junction is usually divided between base and collector, since they are comparably doped.

A Bipolar Junction Transistor (BJT) consists of three differently doped regions. These can have the configuration of npn or pnp and the various layers can either be parallel or perpendicular to the surface. Consider a npn BJT, with three differently doped regions.

1. Emitter region: This is usually a n type heavily doped region. The emitter 'emits' the carriers into the base.
2. Base region: This is a lightly doped p region. The base region is also physically thin so that carriers can pass through with minimal recombination.
3. Collector region: This is a n type region. The collector region has a larger width than the other two regions since charge is accumulated here from the base.

Transistor is a semiconductor device which is used primarily for signal amplification and switching. It is also the fundamental element for integrated circuit design such as the VLSI microprocessor.

Symbol of BJT: Symbols for npn and pnp transistors are as shown in Fig.3.

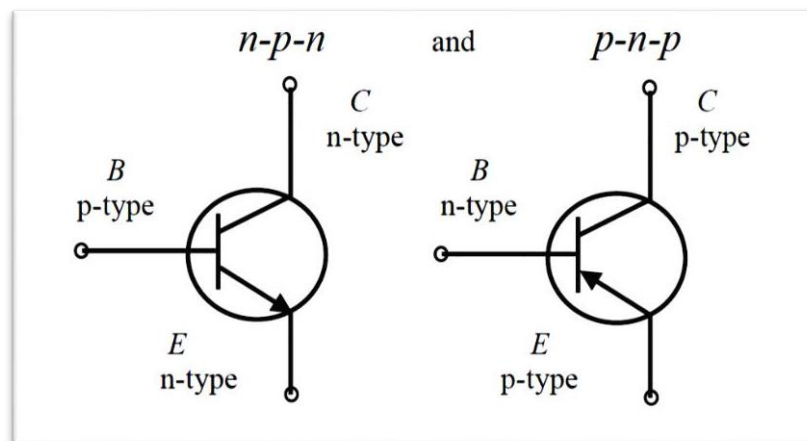


Fig.3: Symbols of npn and pnp transistors

BJT voltages and currents:

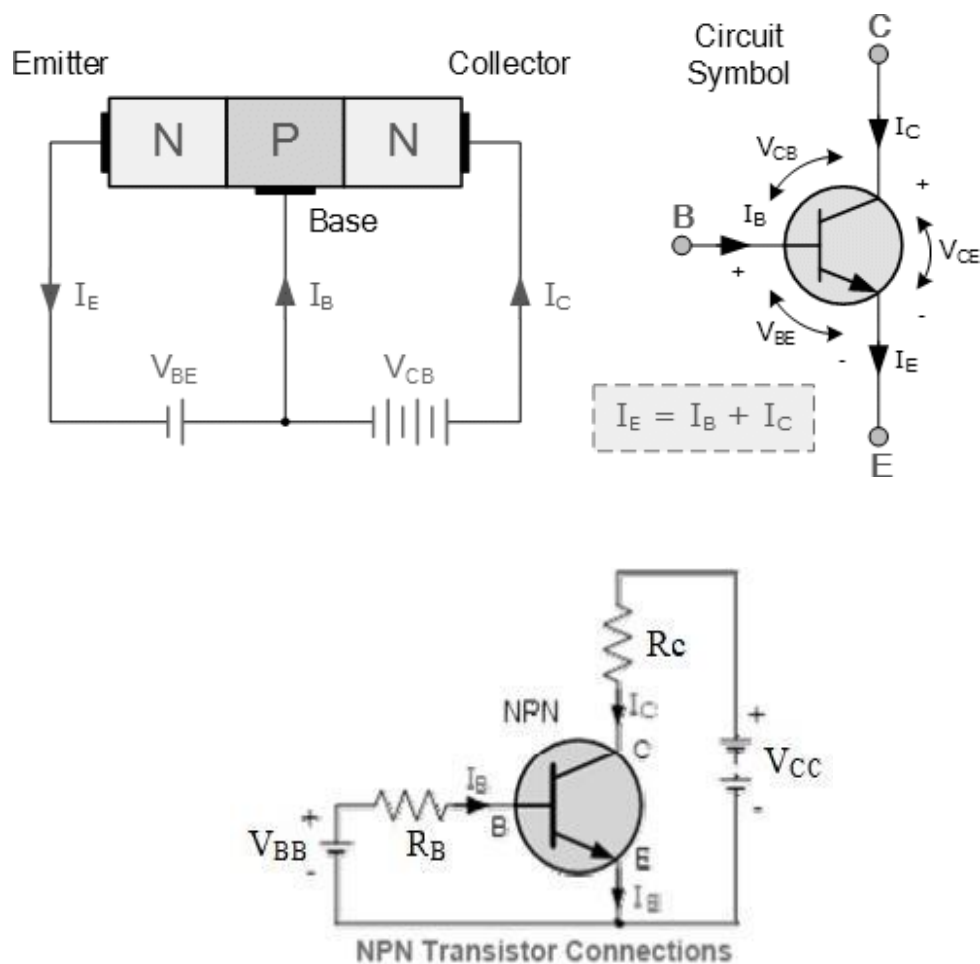


Fig.4: Currents and voltages in a npn transistor

- For npn transistor, base is biased positive with respect to emitter as shown in Fig.4.
- Collector is biased higher positive voltage than base
- Voltage sources are usually connected to the transistor through resistors.
- R_B base resistor and R_C collector resistor are current limiting resistors, V_{CC} collector supply, V_{BB} base voltage
- $V_{CC} > V_{BB}$ ensures collector base junction is reverse biased
- Negative terminals of the supply are connected to emitter
- Typical transistor base emitter voltage, V_{BE} is similar to diode forward voltage 0.7 V for Si transistor and 0.3 V for germanium transistor.
- Collector voltage varies from 3V to 20 V

Transistor Currents:

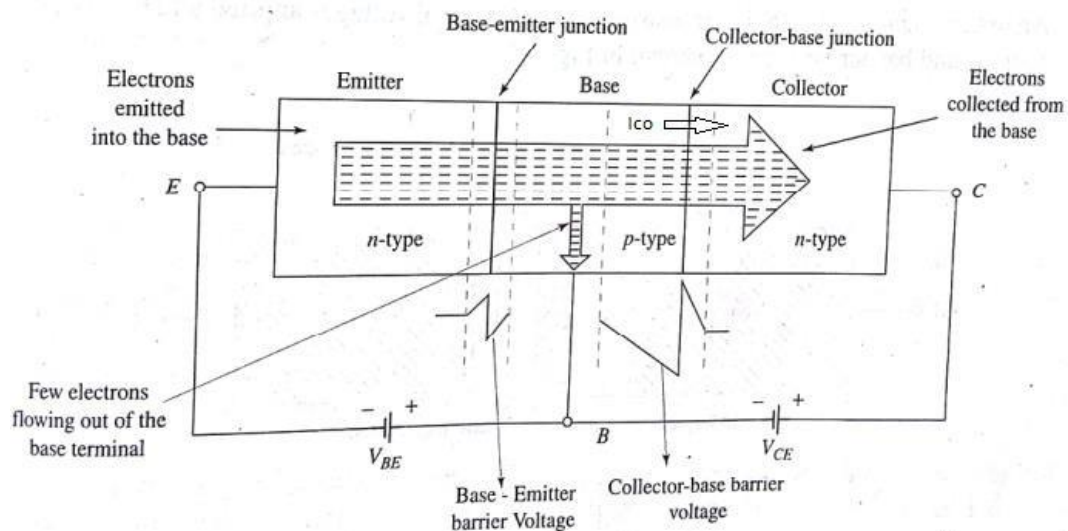


Fig.5: Transistor currents

- The base-emitter junction is forward biased with positive voltage applied to p-type base and negative voltage to n-type emitter.
- This forward bias reduces the barrier voltage enabling the electrons to flow across the junction into the base.
- A large number of these negatively charged electrons are drawn across the collector-base junction by the positive voltage at the collector while a small number of electrons flow out through the base.

As shown in Fig.5, there are three currents flowing in bipolar junction transistor. They are collector current, I_C , emitter current, I_E and base current, I_B . I_C and I_B flow out of the transistor, while I_E flows into the transistor. There is a small portion of the injected carrier recombines with majority carriers in the base to form part of the base current. Thus, the emitter current I_E is equal to $(I_C + I_B)$.

By Kirchhoff's Current Law, KCL, $I_E = I_B + I_C$ (1)

But if we observe in Fig.5, since collector-base junction is reverse biased, there is leakage current, I_{CO} flowing from base to collector along with the current due to majority charge carriers, $I_{C(maj)}$. Leakage current is the current because of minority charge carriers which are drift currents in the collector-base junction. Therefore, total collector current is the sum of leakage current and current due to majority charge carriers.

$$\square \quad I_C = I_{C(maj)} + I_{CBO}$$

The ratio of collector current to emitter current is called α . This parameter is commonly known as **common base gain** as shown in Fig.5.

DC current gain $\alpha = \frac{\text{output current}}{\text{input current}}$

$$\alpha = \frac{I_C}{I_E} \quad (2)$$

The typical value of α_{dc} ranges from 0.95 to 0.99. For a good transistor, its value is closed to one.

The ratio of collector current to base current is β , which also denoted as h parameter- h_{FE} . This parameter is commonly known as **common emitter gain** as shown in Fig.6.

DC current gain $\beta = \frac{\text{output current}}{\text{input current}}$

$$\beta = \frac{I_C}{I_B} \quad (3)$$

The typical range value of β is between 20 to 500.

Substitute equation (1) into equation (3), it yields $I_E = I_B (Q + 1)$

$$I_E = I_C + I_B$$

But $I_C = \alpha_{dc} I_E$

$$\therefore I_C = \alpha_{dc}(I_C + I_B)$$

$$\therefore I_C = \frac{\alpha_{dc}}{1 - \alpha_{dc}} I_B$$

Further $\beta_{dc} = \frac{I_C}{I_B}$

$$\therefore \beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}}$$

$$\beta_{dc} = \frac{\alpha_{dc}}{1 - \alpha_{dc}}$$

or $\beta_{dc}(1 - \alpha_{dc}) = \alpha_{dc}$

$$\beta_{dc} - \beta_{dc}\alpha_{dc} = \alpha_{dc}$$

$$\beta_{dc} = \alpha_{dc} + \beta_{dc}\alpha_{dc}$$

$$= (1 + \beta_{dc})\alpha_{dc}$$

$$\therefore \alpha_{dc} = \frac{\beta_{dc}}{1 + \beta_{dc}}$$

The relationship of α and β parameters shall be $\beta = \frac{\alpha}{1-\alpha}$ **or** $\alpha = \frac{\beta}{1+\beta}$ (4)

Example:

A transistor has $I_B = 0.08\text{mA}$ and $I_E = 9.60\text{mA}$. Determine its collector current I_C , α , and β .

Solution

$$I_C = I_E - I_B = 9.60\text{ mA} - 0.08\text{ mA} = 9.52\text{ mA}$$

$$\alpha = 9.52\text{ mA}/9.60\text{ mA} = 0.9917$$

$$\beta = 9.52\text{ mA}/0.08\text{ mA} = 119\text{or}$$

$$\beta = \alpha/(1-\alpha) = 0.9917/0.0083 = 119$$

For the transistor if $\beta=100$ then corresponding $\alpha = 0.99$.

Transistor Characteristics:

Bipolar Junction Transistor is a three terminal device, there are basically three possible ways to connect it within an electronic circuit with one terminal being common to both the input and output. Each method of connection responding differently to its input signal within a circuit as the static characteristics of the transistor vary with each circuit arrangement.

1. Common Base Configuration - has Voltage Gain but no Current Gain.
2. Common Emitter Configuration - has both Current and Voltage Gain.

3. Common Collector Configuration - has Current Gain but no Voltage Gain.

1. Common Base (CB) Configuration:

As its name suggests, in the **Common Base** or grounded base configuration, the BASE connection is common to both the input signal and the output signal with the input signal being applied between the base and the emitter terminals. The corresponding output signal is measured between the base and the collector terminals as shown in Fig.7.

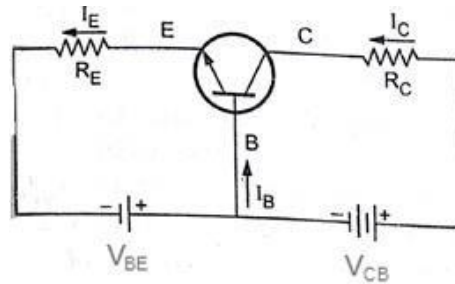


Fig.7: Common base configuration for npn transistor

Input Characteristics:

In this configuration, input is applied between emitter and base terminals of the transistor and output is collected across collector and base terminals. Base is common to input and output terminals.

Input characteristics is the curve between I_E and input voltage V_{EB} for a constant collector base voltage V_{CB} as shown in the Fig.8.

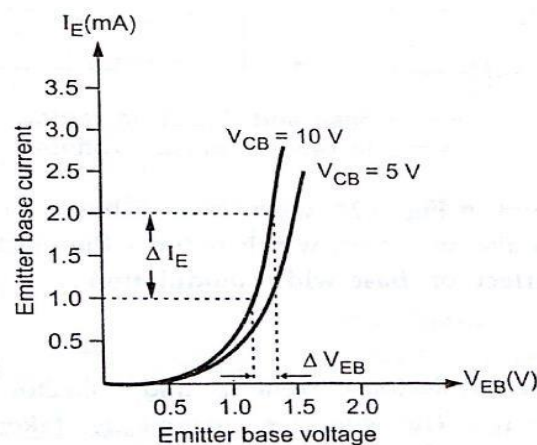


Fig.8: Input characteristics of transistor in CB configuration

After cut in voltage I_E increases drastically with small increase in V_{EB} . Since emitter-base junction is forward biased as shown in Fig.7, the input characteristics are similar to pn-diode forward characteristics.

As V_{CB} increases slight increase in I_E is observed. This is due to change in the width of the depletion region.

Output Characteristics:

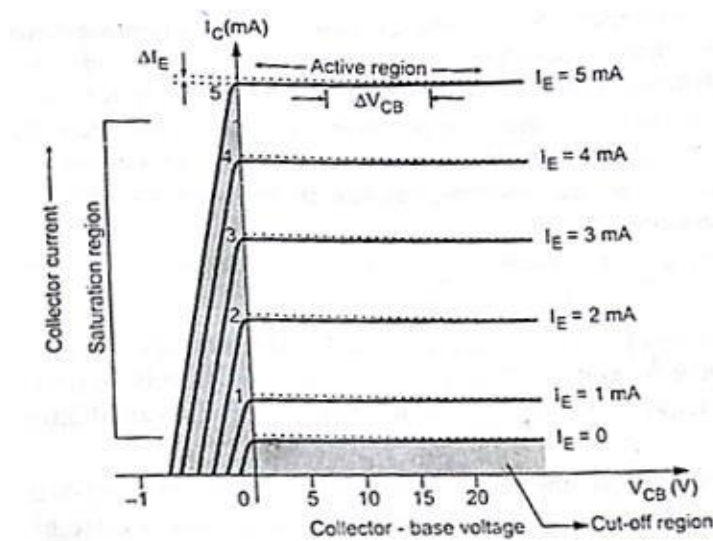


Fig.9: CB output characteristics

It is the curve between collector current, I_C and collector base voltage, V_{CB} for a constant emitter base current, I_E as shown in the Fig.9.

The collector current is along y axis and collector to base voltage is taken along X axis, with emitter current constant.

Three basic regions of operations of transistor can be seen in output characteristics:

State	EB junction	CB junction
Active	Forward Biased	Reverse Biased
Cut off	Reverse Biased	Reverse Biased
saturation	Forward Biased	Forward Biased

Active Region: For the transistor to be operated in the active region, the emitter base junction to be forward biased while the collector base junction to be reverse biased. In this region $I_E \cong I_C$. The transistor works as an amplifier when it is operated in the **active region**. If emitter current is zero, the collector current is just the I_{CBO} .

Cut off region: I_{CBO} is the very small current that appears on the horizontal line as $I_C = 0$. The region below the $I_E=0$ is known as cut off region, where $I_C \cong 0$ where CB and EB junctions are reverse biased.

Saturation Region : Region to the left of $V_{CB} = 0$ V. When V_{CB} is between -1 to 0 volts there is an exponential increase in the collector current. In this region both EB and CB junctions are forward biased.

In the active region the collector current is almost constant and parallel to x axis because $I_E \cong I_C$. The collector current is independent of CB voltage. The transistor is said to be a constant current source. This provides a high dynamic output resistance. As emitter current increases the collector current also increases. The output current, I_C depends on input current I_E , not on collector voltage. The input current controls the output current transistor. Hence transistor is a **current operating device**.

Punch Through Effect:

As shown in Fig.7, as V_{CB} increases the width of the depletion region increases, which reduces the electrical base width. This effect is called as **early effect**. The widened depletion region penetrates into the base until it makes contact with the EB depletion region. This condition is called as **punch through** or reach through effect. When this situation occurs device breakdowns because a large collector current flows through the device which destroys the transistor. Therefore V_{CB} should be kept below $V_{CB(max)}$ to avoid breakdown.

Common Emitter(CE) Configuration:

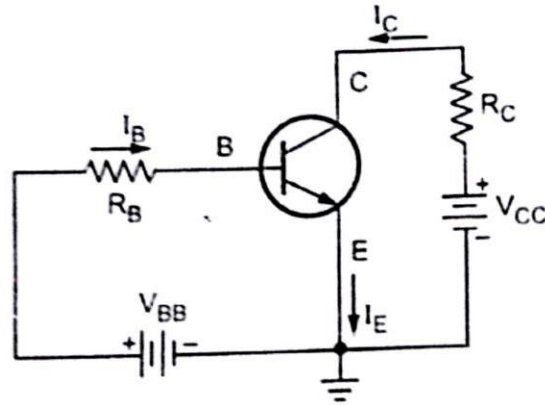


Fig.10: Common Emitter Configuration for npn transistor

As shown in Fig.10, input is applied to base-emitter (BE) terminals and output is collected across collector-emitter(CE) terminals and emitter is common between input and output terminals.

Base voltage, V_{BB} forward biases the BE junction and V_{CC} reverse biases the CB junction.

V_{BE} is the input voltage, V_{CE} is the output voltage, I_B is input current and I_C is output current as shown in figure above.

Input Characteristics:

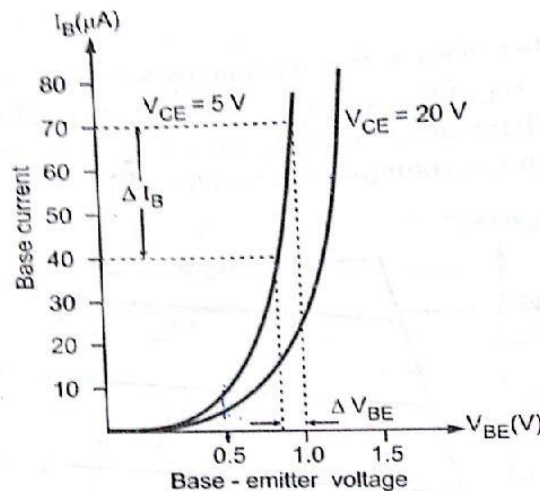


Fig.11: Input characteristics of transistor in CE configuration

It is the curve between input current, I_B and input voltage, V_{BE} at constant collector-emitter voltage, V_{CE} . The base current is taken along y axis and base-emitter voltage is plotted along x axis. Fig.11 shows the input characteristics for CE configuration.

As the input characteristics is between the base to emitter junction, the plot resembles the diode forward characteristics. Since emitter-base junction is forward biased as shown in Fig.10, the input characteristics are similar to pn-diode forward characteristics. After the cut in voltage, the I_B increases rapidly with small increase in voltage, V_{BE} . The dynamic input resistance is small in CE configuration. It is the ratio of change in base current at constant collector emitter voltage.

For a fixed value of V_{BE} , base current I_B decreases as V_{CE} increased. A larger value of V_{CE} results in a large reverse bias at collector base pn junction. This increases the depletion region and reduces effective base width. Due to very few recombinations in the base region base current, I_B reduces.

Output Characteristics:

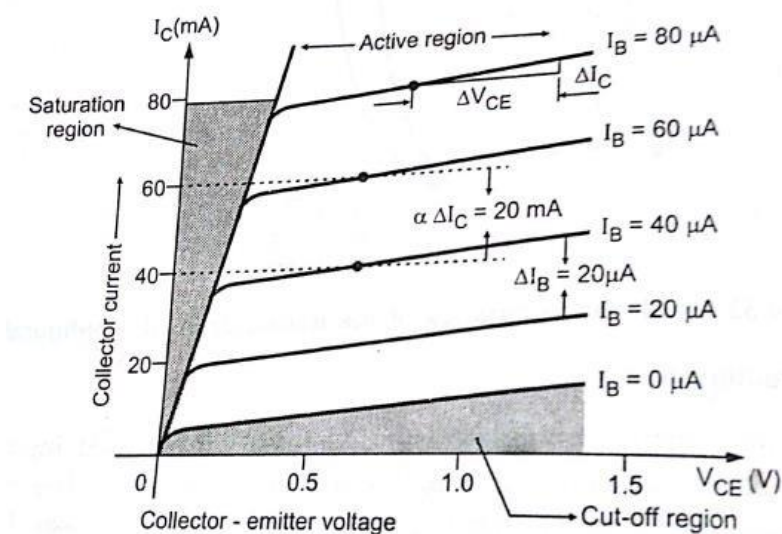


Fig.12: Output characteristics for CE configuration

Output characteristics display the relation between collector current, I_C and collector voltage, V_{CE} for different values of I_B as shown in Fig.12. These characteristics are also called as collector characteristics.

The output characteristics of common emitter configuration has three basic regions of operations: Active, Cutoff, Saturation regions.

Active region: In this region curves are approximately horizontal. Collector base junction is reverse biased. As V_{CE} increases the reverse bias increases and the depletion region widens. This reduces the chances of recombination at the base which will increase α_{dc} . Due to this early effect collector current increases more sharply.

Saturation region: If V_{CE} reduced to a small value such as 0.2 volts, then CB junction is forward biased and EB junction is already forward biased by 0.7 volts. Both the junctions are forward biased and transistor operates in the saturation region. The saturation value of V_{CE} reduced to a small value such as 0.2 volts, then CB junction is forward biased and EB junction is already forward biased by 0.7 volts. Both the junctions are forward biased and transistor operates in the saturation region. The saturation value of V_{CE} denoted as $V_{CE(max)}$ ranges between 0.1 volt to 0.3 volts.

Cutoff region: when the input base current is made equal to zero, reverse leakage current flows I_{CEO} . Region below $I_B = 0$ is the cutoff region, where both the junctions are reverse biased.

DC Load line Analysis:

A DC load line is a straight line drawn on the output CE characteristics of a transistor circuit that describes all the DC conditions that exist in the operation of a circuit. Consider npn transistor in CE configuration biased in the active region as in the Fig.13.

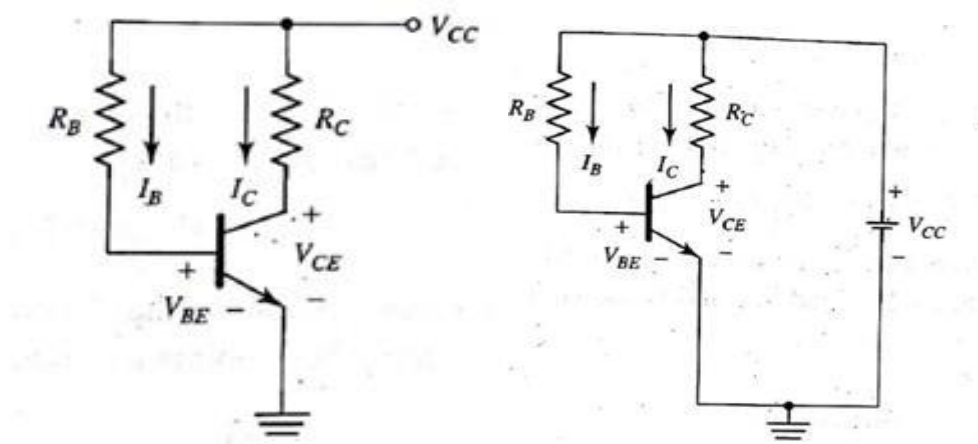


Fig.13: Transistor in CE configuration

The supply voltage, V_{CC} forward biases the base-emitter junction. Applying KVL to base-emitter loop, we get

$$\begin{aligned} V_{CC} &= I_B R_B + V_{BE} \\ \Rightarrow I_B &= \frac{V_{CC} - V_{BE}}{R_B} \end{aligned} \quad (1)$$

And we know that $I_C = \beta I_B$ (2)

Applying KVL to collector- emitter loop, we have

$$V_{CC} = I_C R_C + V_{CE} \quad (3)$$

When $I_C = 0$, from eq.(3), we get $V_{CC} = V_{CE}$.

The transistor is said to be at cut-off region.

When $V_{CE} = 0$, from eq.(3), we get

$$\begin{aligned} I_C R_C &= V_{CC} \\ \text{or } I_C &= \frac{V_{CC}}{R_C} \end{aligned}$$

Now the transistor is said to be in saturation. The corresponding points are plotted on CE output characteristics of the transistor as shown in Fig.14. The straight line obtained by joining these two points is called as DC load line.

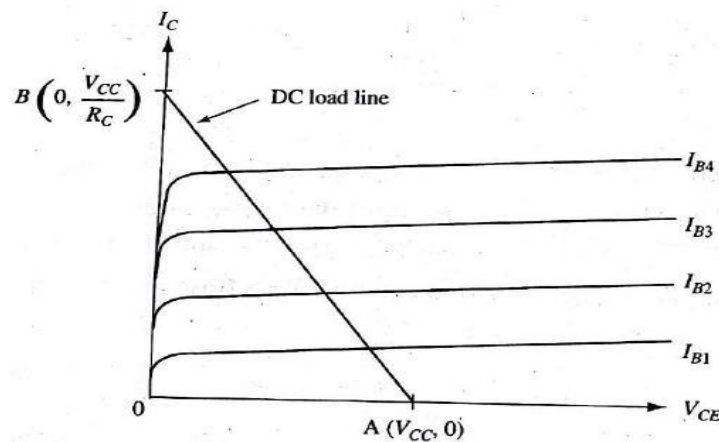


Fig.14: DC load line drawn on transistor CE output characteristics

Applications of Transistor: Transistor as a Switch:

BJT can be used as a switch when its operating point is moved from saturation to cutoff region. Fig.15 shows connection diagram.

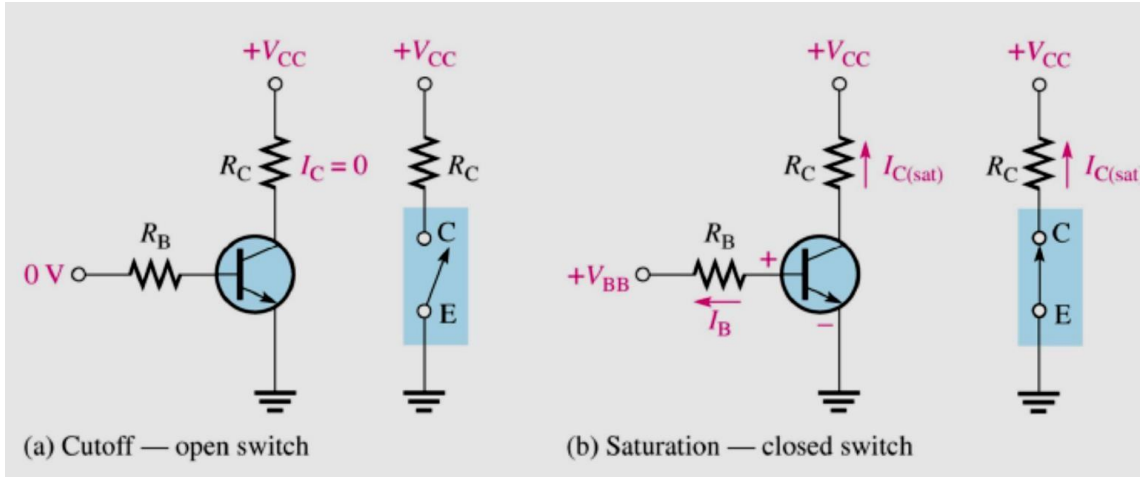


Fig.15: Transistor as a switch

When V_{BB} is zero, corresponding I_B is also zero. Since there is no base current, transistor will be OFF. The operating point of the transistor will be in the cut-off region. When transistor is OFF, I_C is also zero. This corresponds to the open switch situation.

When V_{BB} is positive, I_B flows through the base and transistor turns ON. The operating point is shifted to saturation region as both junctions are forward biased. When transistor is ON, this corresponds to a closed switch condition. Fig.16 shows the operating regions when transistor used as switch.

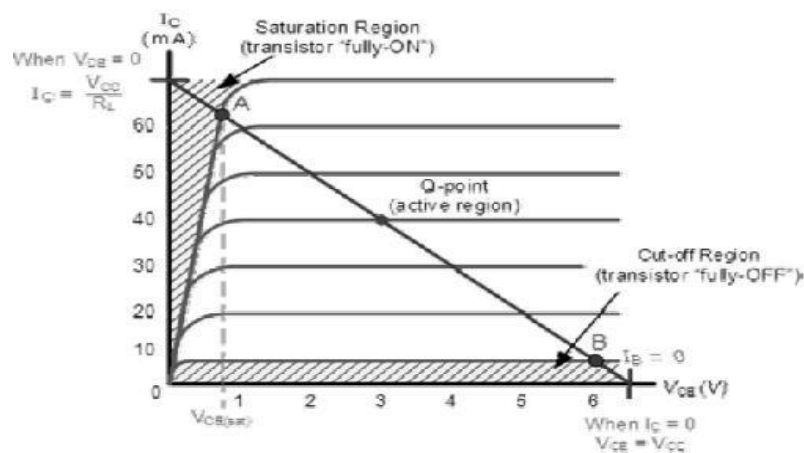


Fig.16: Transistor as a switch operating regions

A Simple Application of a Transistor Switch:

A transistor used to switch an LED on and off. The transistor in Fig.17 is used as a switch to turn the LED on and off. For example, a square wave input voltage with a period of 2 seconds is applied to the input as indicated. When the square wave is at 0 V, the transistor is in cutoff and since there is no collector current, the LED does not emit light. When the square wave goes to its high level, the transistor saturates. This forward-bias the LED, and the resulting collector current through the LED causes it to emit light. Thus, the LED is on for 1 second and off for 1 second.

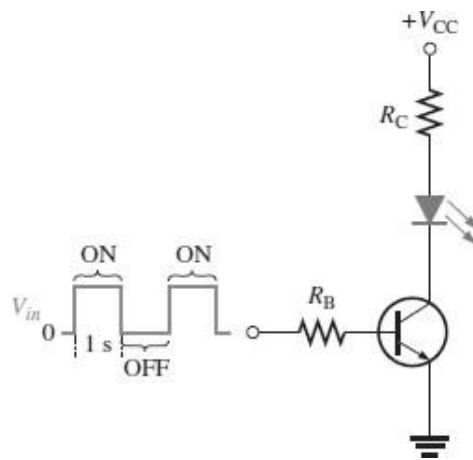


Fig.17: A transistor used to switch an LED on and off

Transistor as an amplifier:

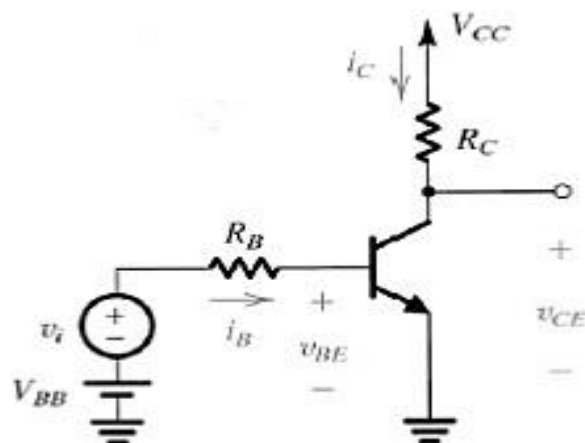


Fig.18: Transistor as an amplifier

A simple circuit diagram for transistor used as an amplifier is as shown in Fig.18. The operating point for the transistor to work as an amplifier should be at the centre of the active region.

We know that $I_C = \beta I_B$ and $I_C = \alpha I_E$ and also $I_C \cong I_E$.

Input voltage V_{in} is superimposed on the dc bias voltage V_{BB} . Without ac, when only DC voltages are applied, $V_C = V_{CE} = V_{CC} - I_C R_C$.

When ac signal is applied, $V_C \pm v_c = V_{CE} = V_{CC} - (I_C \pm i_c) R_C$

As input voltage V_{in} increases the collector current I_C increases and V_{out} decreases. On the other hand, as V_{in} decreases the collector current decreases and hence output voltage increases. Therefore, at the output of the amplifier there will be 180° phase shift observed.

Single stage RC coupled CE amplifier:

RC coupled single stage common emitter audio frequency amplifier is as shown Fig.19. Resistors R_1 , R_2 , R_E form voltage divider biasing circuit for the amplifier. This will set the operating point in the active region for the CE amplifier. The input capacitor C_1 is couples the signal to the base of the transistor. This capacitor blocks any dc component present in the signal and passes only the ac signal. Thereby C_1 helps in fixing the operating point in the active region.

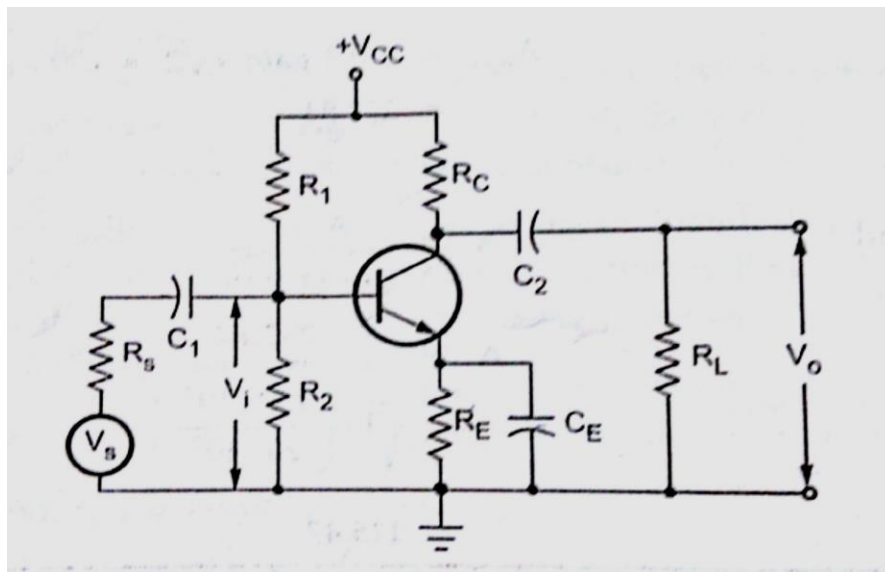


Fig.19: Single stage RC coupled CE amplifier

The capacitor C_E is the emitter bypass capacitor connected in parallel to the emitter resistance R_E to provide a low reactance path for the amplified signal. Without this resistor there will be voltage drop across the R_E which will reduce the output voltage, thereby changing the gain of the amplifier.

The output capacitor C_2 is used as coupling capacitor to couple the output of the amplifier to the load or to the next stage. C_2 blocks dc and passes only ac signal.

The source resistance R_S is in parallel to the R_2 will reduce the bias voltage at the transistor base and intern it reduces the collector current. Similarly, when R_L is connected directly the dc levels of V_C and V_{CE} will change. To avoid this and to maintain stability of bias condition the coupling capacitors are used.

Phase reversal:

During positive half cycle, two voltages ac and dc will be added increasing the forward bias on base emitter junction. This increases the base current. The collector current is β times of base current, hence collector current will also increase. This increases the voltage drop across R_C . Since $V_C = V_{CC} - I_C R_C$ the increase in I_C results in a drop in the V_C as V_{CC} is constant. Thus, as V_i increases in the positive direction V_o goes in the opposite direction and we get negative half cycle of the output voltage as shown in Fig.20.

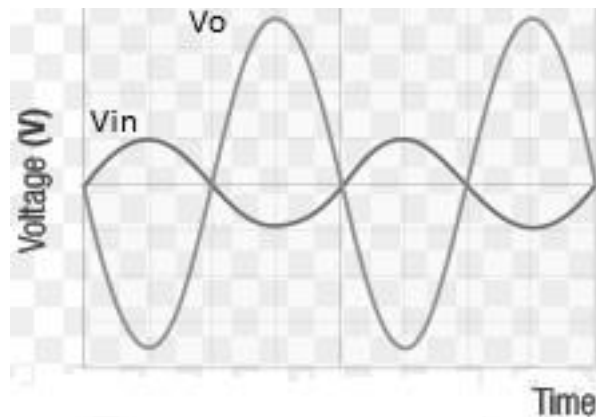


Fig.20: Input/output waveforms of RC coupled amplifier

During negative half cycle of input, ac and dc voltage oppose each other reducing the forward bias on the base emitter pn junction. This reduces the base current. Accordingly, collector current and inter drop across collector resistor reduce, increasing the output voltage. Thus, we

get a positive half cycle at the output for the negative half cycle of the input. Hence 180° phase shift is observed between the input and output voltages for a CE amplifier.